## Aluminum 7072-H113

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| **Categories:** | Metal; Nonferrous Metal; Aluminum Alloy; 7000 Series Aluminum Alloy |
| **Material Notes:** | Data points with the AA note have been provided by the Aluminum Association, Inc. and are NOT FOR DESIGN.<br><br>**Composition Notes**:<br>Composition information provided by the Aluminum Association and is not for design. |
| **Key Words:** | UNS A97072; ISO AlZn1; Aluminium 7072-H113; AA7072-H113; Al7072-H113 |
| **Vendors:** | No vendors are listed for this material. Please click here if you are a supplier and would like information on how to add your listing to this material. |

| Physical Properties | Metric | English | Comments |
|---|---|---|---|
| Density | 2.72 g/cc | 0.0983 lb/in³ | AA; Typical |

| Mechanical Properties | Metric | English | Comments |
|---|---|---|---|
| Hardness, Brinell | 21 | 21 | 500 kg load with 10 mm ball. Calculated value. |
| Tensile Strength, Ultimate | 75.0 MPa | 10900 psi | |
| Elongation at Break | 15 %<br>@Thickness 1.60 mm | 15 %<br>@Thickness 0.0630 in | In 5 cm |
| Tensile Modulus | 68.0 GPa | 9860 ksi | |
| Compressive Modulus | 70.0 GPa | 10200 ksi | |
| Poissons Ratio | 0.33 | 0.33 | Estimated from trends in similar Al alloys. |
| Shear Modulus | 26.0 GPa | 3770 ksi | Estimated from similar Al alloys. |
| Shear Strength | 50.0 MPa | 7250 psi | Calculated value. |

| Electrical Properties | Metric | English | Comments |
|---|---|---|---|
| Electrical Resistivity | 0.00000290 ohm-cm | 0.00000290 ohm-cm | |

| Thermal Properties | Metric | English | Comments |
|---|---|---|---|
| CTE, linear | 21.8 µm/m-°C<br>@Temperature -50.0 - 20.0 °C | 12.1 µin/in-°F<br>@Temperature -58.0 - 68.0 °F | |
| | 23.6 µm/m-°C<br>@Temperature 20.0 - 100 °C | 13.1 µin/in-°F<br>@Temperature 68.0 - 212 °F | AA; Typical; average over range |
| | 24.5 µm/m-°C<br>@Temperature 20.0 - 200 °C | 13.6 µin/in-°F<br>@Temperature 68.0 - 392 °F | |
| | 25.5 µm/m-°C<br>@Temperature 20.0 - 300 °C | 14.2 µin/in-°F<br>@Temperature 68.0 - 572 °F | |
| Specific Heat Capacity | 0.893 J/g-°C | 0.213 BTU/lb-°F | |
| Thermal Conductivity | 222 W/m-K | 1540 BTU-in/hr-ft²-°F | |
| Melting Point | 640.6 - 657.2 °C | 1185 - 1215 °F | AA; Typical range based on typical composition for wrought products >= 1/4 in. thickness |
| Solidus | 640.6 °C | 1185 °F | AA; Typical |
| Liquidus | 657.2 °C | 1215 °F | AA; Typical |

| Processing Properties | Metric | English | Comments |
|---|---|---|---|
| Annealing Temperature | 343 °C | 650 °F | |

| Component Elements Properties | Metric | English | Comments |
|---|---|---|---|
| Aluminum, Al | <= 97.5 % | <= 97.5 % | As remainder |
| Copper, Cu | <= 0.10 % | <= 0.10 % | |
| Magnesium, Mg | <= 0.10 % | <= 0.10 % | |
| Manganese, Mn | <= 0.10 % | <= 0.10 % | |
| Other, each | <= 0.05 % | <= 0.05 % | |
| Other, total | <= 0.15 % | <= 0.15 % | |
| Si+Fe | <= 0.70 % | <= 0.70 % | |
| Zinc, Zn | 0.80 - 1.3 % | 0.80 - 1.3 % | |

**References** for this datasheet.